

WASP-50b

R-1651

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-50_140131 · created 2026-08-02T08:13:09+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456688.654 +/- 0.085 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.154 +/- 0.05
Transit depth (Rp/Rs)^2	2.39 +/- 1.55 [%]
Orbital Inclination (inc)	89.74 +/- 2.91
Airmass coefficient 1 (a1)	1.004 +/- 0.037
Airmass coefficient 2 (a2)	0.003 +/- 0.014
Scatter in the residuals of the lightcurve fit is	3.71 %
Best Comparison Star	#3 - [341, 220]
Optimal Aperture	2.49
Optimal Annulus	10.29
Transit Duration (day)	0.056 +/- 0.035

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.154 $\pm$ 0.05 vs 0.1404 (deviation 0.27 σ)
Duration fit vs published	0.056 d vs 0.0754 d (deviation 0.55 σ)
Mid-time O–C	-3.4 min
Depth SNR	3.1
Residual scatter	3.71 %

Light curve

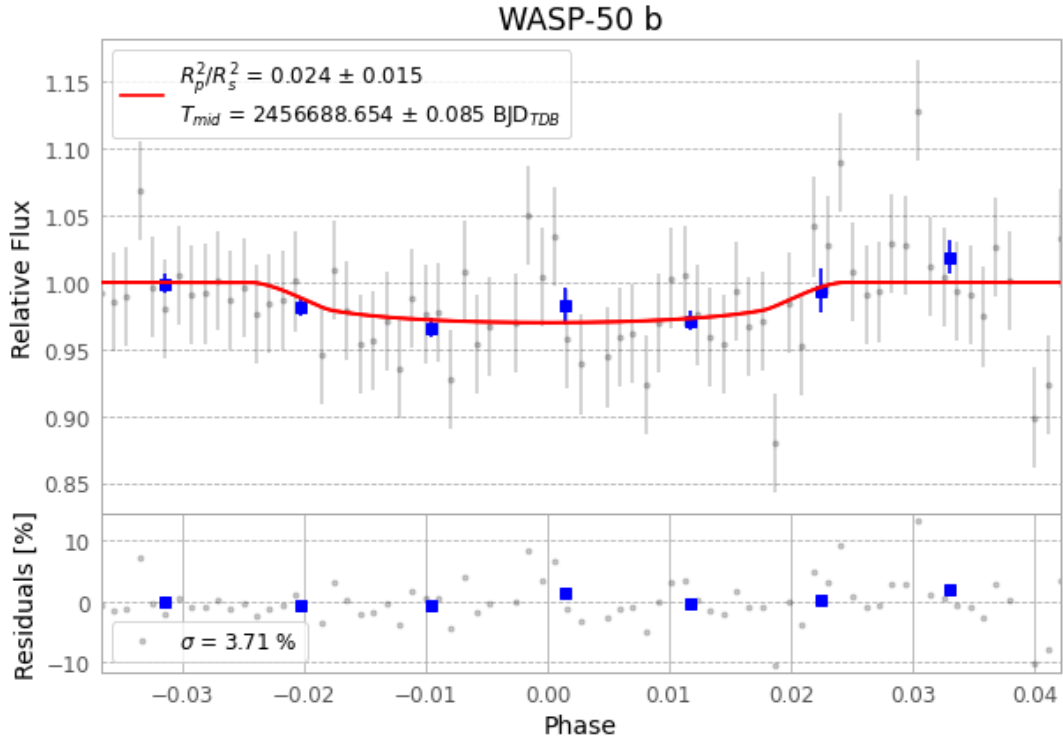


Figure 1 — FinalLightCurve_WASP-50 b_2014-01-30.png (SHA-256 9491db381ff69ae3...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [353, 242], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0c2e7f80c286aa60...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

76 FITS frames (50 MB), WASP-50140131015712.FITS → WASP-50140131054212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-140131.log (a3bbef915cb3...), FinalLightCurve_WASP-50 b_2014-01-30.png (9491db381ff6...), FinalLightCurve_WASP-50 b_2014-01-30.csv (b3f240d18505...)

Compute resources

Wall time 185.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 08:13Z.